

INTRODUCTION:

The rapid adoption of electric vehicles (EVs) has necessitated the development of efficient battery management and swapping systems. As EVs become more prevalent, the need for convenient and user-friendly battery swapping solutions has become paramount. This project, named "Energy Tag," aims to streamline the process of managing and pricing EV batteries in battery swap stations. By leveraging advanced technologies such as RFID (Radio Frequency Identification), voltage sensing, and microcontroller-based automation, this project seeks to enhance the efficiency and user experience in battery swap stations. Battery swapping stations offer a significant advantage over traditional charging methods by allowing EV users to quickly exchange their depleted batteries for fully charged ones. This reduces the downtime associated with charging, making it an attractive solution for fleet operators and heavy-duty vehicle users where time is a critical factor. Despite the advantages, current systems face challenges in user identification, slot management, and accurate billing based on battery usage. Fast charging station infrastructure that is both effective and responsive to user's needs, plays a vital role in electric vehicles (EVs) progress. The idea of the dynamic pricing of batteries in the battery swapping stations is a quite revolutionary form of the implementation of such endeavor. Through the RFID integration in EV batteries and the application of real-time data analytics, the proposed method of managing EV batteries and pricing will radically differ from the already applied approaches. In this regard, the system integrates RFID technology with EV batteries and hence operates automatically with operation parameters being retrieved such as charging, discharging, and load and aging. Through scanning of the RFID-tagged battery at swapping stations real-time data (parameters) are extracted, which in turn helps in knowing the current situation and the status of the batter.

Having this knowledge at hand, the system applies model algorithms for understanding the optimal price for battery replacement. The pay-per-usage model is moldable and reactive with respect to qualities like battery state, demand, and network congestion. The calculated price on displayed LCD screens gives EV owners rooted idea about the swapping during the process. This creates clarity and transparency for them as they go through the swapping process. Generally, DPM in BSS can be recognized as a revolutionary change of EV charging infrastructure. Through using RFID technology, real-time data analytics, and transparent pricing mechanism this unusual approach provides user experience, puts into practice of sustainable battery management, which in its turn drives the paving of the way to a cleaner and more effective transportation ecosystem. In addition, it is embedded not only to meet resource allocation and user engagement needs but also to promote social sustainability by encouraging responsible battery behavior among users

Energy Tag embodies the convergence of sustainable transportation and cutting-edge technology, underscoring the critical role of innovation in the green energy sector. As the world grapples with climate change and the imperative to reduce carbon emissions, efficient and accessible EV infrastructure becomes essential. This project showcases how technological advancements can be harnessed to support environmental goals, promoting a cleaner and greener future. By streamlining the battery swap process, Energy Tag not only alleviates the challenges faced by EV users but also exemplifies the potential for scalable

LITERATURE SURVEY SUMMARY:

Paper that are studied in literature survey provides an overview of lithium-ion batteries and Battery Management Systems (BMS) in Electric Vehicles (EVs), focusing on safety, voltage control, temperature regulation, and battery longevity. It details BMS functions such as parameter determination, state estimation, failure assessment, safe control, charge management, heat management, data communication, and memory. The paper outlines two State of Charge (SOC) estimation methods and four battery management techniques: CC, CV, CC-CV, and MCC charging. It explores computational intelligence techniques for battery management and discusses three approaches to lithium battery modeling: Black-Box, Electrical Lumped-Model, and Electrochemical System Modeling. Circuit protection features and techniques like Cell/Module Bypass and Shared Cell-to-Cell balancing are analyzed. The methodology includes a descriptive and comparative analysis of three battery swapping modules (BSM, ISM, ASM), a literature review, case studies of Tesla Motors and NIO Inc., and an examination of Power Electronics Converters (PEC) used in EV manufacturing, emphasizing DC-DC converters and inverters for motor operation.

PROBLEM STATEMENT:

Efficiently managing EV battery swapping is challenging due to issues with battery longevity, safety, and accurate SOC estimation. This project aims to enhance battery swapping by improving monitoring, booking systems, and real-time data communication.

OBJECTIVES:

1. To develop a reliable battery swapping system that ensures safety, efficiency, and longevity of lithium-ion batteries.
2. To implement real-time monitoring and data communication for accurate SOC estimation and battery management.
3. To streamline the battery booking and swapping process for improved user experience and operational efficiency.

DESIGN PROCESS:

1. Empathize:

Understanding the needs of EV users at battery swap stations was crucial for the project. Common issues identified include:

- Long wait times during battery swaps.
- Lack of transparency in pricing and billing.
- Uncertainty about the health and charge level of swapped batteries.

2. Define :

To address these issues, the core problems were defined as follows:

- Problem Statement: EV users experience inefficiencies, unclear pricing, and reliability concerns during battery swapping.
- User Personas: Typical users are EV owners who require fast, reliable, and cost-effective battery swapping services.
- Journey Maps: The user experience involves arriving at the station, swapping the battery, and paying for the service. Key pain points include waiting time and pricing transparency.

3. Ideate:

Several innovative solutions were brainstormed to solve the defined problems:

- Automated Booking System: Implementing a web interface for booking swap slots in advance to reduce wait times.
- Transparent Pricing Mechanism: Developing a system that calculates and displays the cost based on battery charge levels in real-time.
- Battery Health Monitoring: Using sensors to measure and display the battery's health and charge level before and after the swap.

4. Prototype:

Prototypes were developed to test the feasibility of the proposed solutions:

- Hardware Setup: Included an RFID reader, voltage sensors, LCD display, and relays to manage battery swaps and display information.
- Software Application: Developed to control the hardware components, process user data, and communicate with the server.
- Web Interface: Created for users to book slots, view battery status, and understand pricing.

5. Test:

The prototypes were rigorously tested to ensure functionality and user satisfaction.

HARDWARE USED:

1. **ESP32:** ESP32 is a low-cost, low-power microcontroller (MCU) with integrated Wi-Fi and Bluetooth connectivity. It's designed for mobile devices, wearable electronics, and IoT applications, and is supported by the Arduino platform

In this specific project, the ESP32 is utilized for:

- Connecting to Wi-Fi: To access the internet for updating Google Sheets and hosting the web server.
- Reading RFID Data: Using SPI communication to interact with the RFID reader.
- Driving the LCD Display: Using I2C communication to show messages to the user.
- Handling Web Requests: Running a web

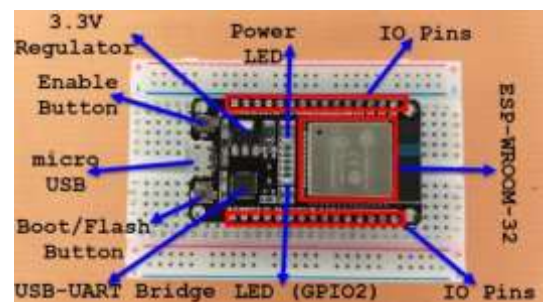


Figure 1 ESP 32

server to allow booking slots via a web interface.

- Controlling the Relay: Managing the power to the batteries during swap and measurement operations.
- Measuring Battery Voltage: Using ADC to read the voltage from the battery voltage dividers to determine the battery charge percentage

2. MFRC522 RFID module :

Radio Frequency Identification (RFID) technology uses electromagnetic fields to automatically identify and track tags attached to objects. An RFID system consists of a tag, an antenna, and a reader. The tag contains a microchip that stores information, while the reader captures this information through the antenna. RFID technology is widely used for its ability to read multiple tags simultaneously, its non-line-of-sight nature, and its durability in various environments.



Figure 2 RFID

3.LCD display:

A 16×2 LCD display is a liquid crystal display that can show 16 characters in each of its two rows, providing a total of 32 characters of information. It's commonly used to display alphanumeric information in various electronic devices.



Figure 3 LCD

4.Battery:

Lithium-ion is the most popular rechargeable battery chemistry used today. Lithium-ion batteries power the devices we use every day, like our mobile phones and electric vehicles. Lithium-ion batteries consist of single or multiple lithium-ion cells, along with a protective circuit board.



Figure 4 BATTERY

5.2Channel Relay Module:

The 2-channel 5V relay module is an essential component for safely interfacing the low- power control signals from the ESP32 with high-power



Figure 5 2 channel relay module

devices. It provides isolation and reliable switching, making it ideal for use in applications like a battery swap station where controlling high-power charging circuits is crucial.

In our battery swap station project, a 2-channel relay can be used to:

1. Control Power to Battery Chargers:

- Ensure that batteries are only charged when a valid user is detected and the slot is available.
- Safely disconnect the charger once the swapping or charging process is complete.

2. Control Other High-Power Devices:

- Potentially control other peripherals like cooling fans, indicator lights, or any device requiring higher voltage and current than the ESP32 can directly handle.

6. Battery Management System (BMS):

A Battery Management System (BMS), which manages the electronics of a rechargeable battery, whether a cell or a battery pack, thus becomes a crucial factor in ensuring electric vehicle safety.



Figure 6 BMS

It safeguards both the user and the battery by ensuring that the cell operates within its safe operating parameters. BMS monitors the State Of Health (SOH) of the battery, collects data, controls environmental factors that affect the cell, and balances them to ensure the same voltage across cells.

Integrating a BMS into our battery swap station ensures safe operation, efficient battery management, and enhanced user experience. The ESP32 can communicate with the BMS to monitor battery status and control charging/discharging processes, ensuring the system runs smoothly and safely.

7. Motor Driver:

The Maker Drive is a robust and flexible motor driver module that can significantly enhance your



Figure 7 Maker Driver

projects by providing precise control over DC motors. Its integration with the ESP32 allows for sophisticated motor control applications, making it suitable for automated systems in your battery swap station.

In our battery swap station project, the Maker Drive could be used for:

1. **Automated Mechanical Systems:**
 - Controlling motorized mechanisms for battery swapping.
 - Moving robotic arms or conveyor belts to position batteries.
2. **Cooling Systems:**
 - Controlling fans for cooling batteries during charging/discharging.
3. **Indicator Systems:**
 - Using small motors to trigger mechanical indicators showing the status of slots (e.g., available, booked, in use).

8. Voltage Sensor:

The voltage sensor is a crucial component in your battery swap station project because it allows you to measure the battery's charge level accurately. This measurement is necessary for billing, safety, and providing feedback to users. The ESP32 is an excellent choice for this project due to its WiFi capability, sufficient I/O pins, ADC features, and processing power, enabling it to handle various tasks required in this complex system.



Figure 8 Voltage Sensor

SOFTWARES USED:

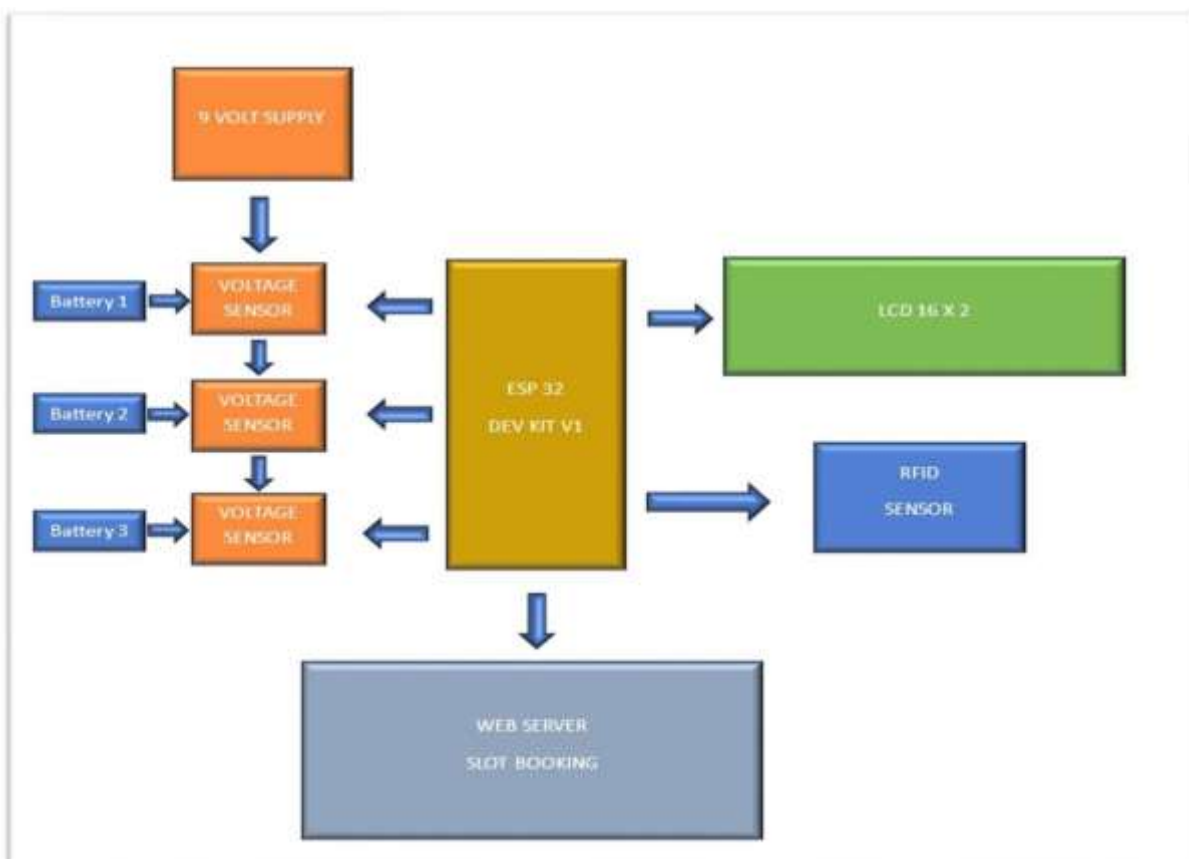
In this project we have used Arduino IDE for coding.

The Arduino Integrated Development Environment (IDE) is a cross-platform application used for writing, compiling, and uploading code to Arduino and compatible microcontroller boards. It provides an intuitive interface and a comprehensive set of tools that simplify the process of programming microcontrollers. The Arduino IDE supports a wide range of microcontrollers, including the ESP32 used in the "Energy Tag" project.

NOVELTY:

1. Integration of RFID technology for secure and seamless user identification during battery swaps.
2. Real-time battery monitoring using voltage sensors to ensure accurate SOC estimation and optimal charging management.
3. Automated slot booking system via a web server, enhancing user convenience and system efficiency.
4. Implementation of Google Sheets API for tracking and recording battery usage and user data for transparent and reliable documentation.
5. Use of an ESP32 microcontroller for advanced connectivity and processing capabilities, enabling efficient system operation and data communication.

BLOCK DIAGRAM:



METHODOLOGY:

The methodology for the "Energy Tag - Streamlining EV Battery Prices in Battery Swap Stations" project involves a comprehensive approach combining hardware setup, software integration, and user interaction. Initially, the hardware components are configured, including an LCD display for user interface, an RFID reader to identify users, and WiFi connectivity using the ESP32's built-in capabilities. Relays and voltage sensors are employed to control power flow and monitor battery voltage levels, respectively.

In the setup phase, serial communication is established for debugging, and the LCD, RFID reader, and WiFi connection are initialized. The ESP32 connects to a specified network and displays its IP address on the LCD for user reference. The relay and web server are also initialized to handle client requests and control power distribution to the batteries.

The RFID system continuously scans for new cards, identifying users and managing their interactions with the battery slots. When a user returns a battery, the system measures the battery voltage to determine its state of charge, calculates the price based on the remaining charge, and processes the refund. If a user is booking a new slot, the system records their details and the initial charge of the battery, calculating the cost based on the initial charge percentage.

The web server provides a user-friendly interface for booking battery slots, allowing users to view available slots and submit their booking details through an HTML form. Data about each transaction, including user information, slot number, battery charge, and pricing, is sent to a Google Sheet for record-keeping and analysis. The LCD display provides real-time feedback to users, confirming bookings, showing refund amounts, and displaying status updates. This integrated approach ensures efficient management of battery swapping, accurate transaction logging, and user-friendly operation, making the battery swap station a streamlined solution for EV battery management.



Figure 9 Model

RESULTS:



Figure 10 IP address display

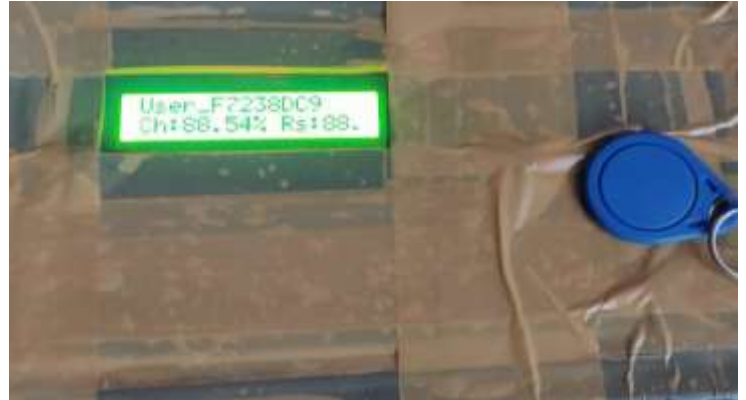
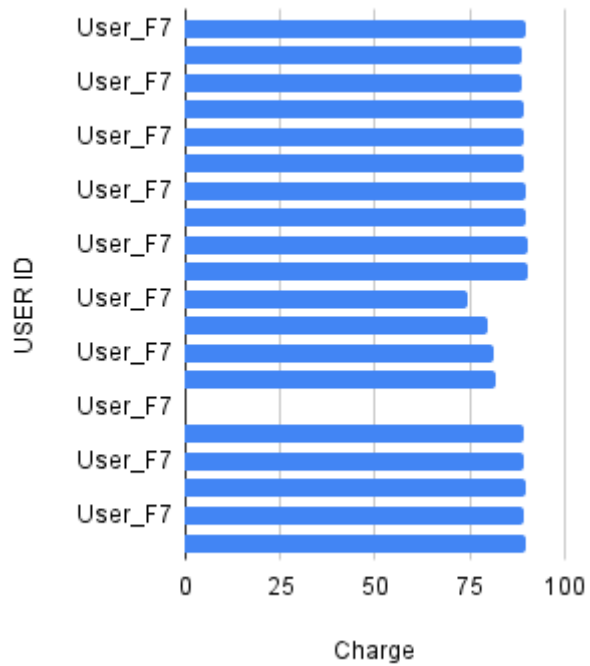


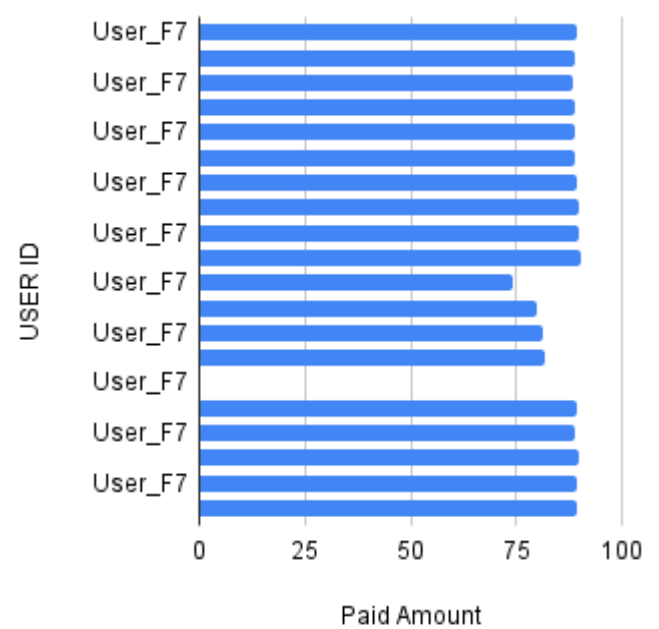
Figure 11 User ID and Amount Display

Charge



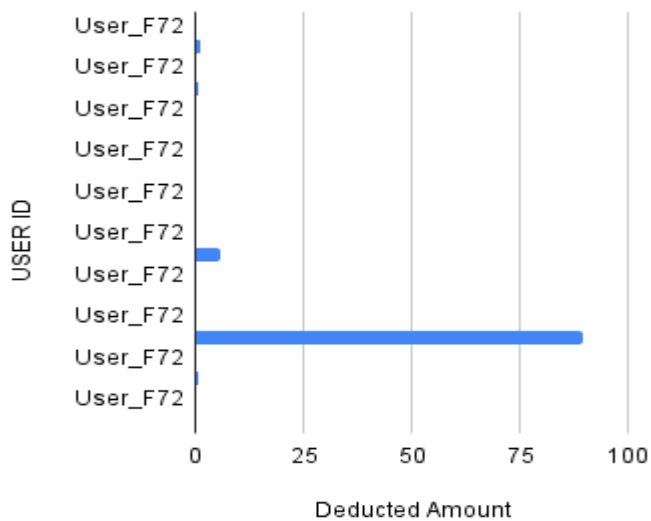
Graph 1

Amount in the wallet



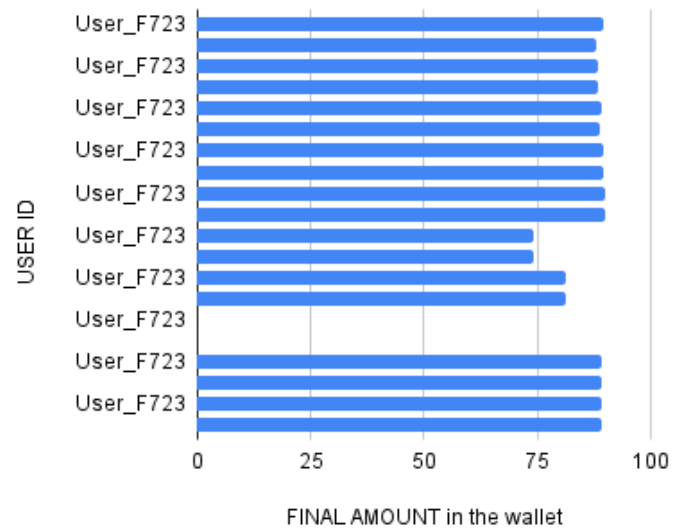
Graph 2

Deducted Amount



Graph 3

FINAL Amount in the wallet



Graph 4

FUTURE SCOPE:

In the future, the project aims to advance battery monitoring with techniques like Coulomb counting and impedance spectroscopy for accurate state-of-charge estimation. Additionally, enhancing the user interface through mobile apps or web portals can improve the overall experience. Remote management capabilities will enable administrators to monitor battery status and usage remotely, with analytics offering insights for optimization. Integration of IoT technology can facilitate proactive maintenance and fault detection. Further, collaboration with EV manufacturers for standardization and expansion to multiple locations will promote widespread adoption and sustainability.

CONCLUSION:

The "Energy Tag - Streamlining EV Battery Prices in Battery Swap Stations" project successfully integrates hardware and software components to create an efficient, user-friendly battery swap station. By leveraging the ESP32's capabilities, RFID technology, and WiFi connectivity, the system ensures accurate user identification, real-time transaction processing, and seamless data management. The use of LCD displays enhances the user interface, providing clear instructions and status updates. Voltage sensors and relays efficiently manage the battery charging and swapping processes, ensuring safe and reliable operation. The implementation of a web server allows for convenient slot booking and monitoring, enhancing user experience and operational efficiency. The integration with Google Sheets for data logging ensures accurate record-keeping and facilitates analysis for future improvements. Overall, the project demonstrates a robust solution for managing EV battery swaps, providing a scalable model that can be expanded and adapted for broader applications in the EV charging infrastructure. This project paves the way for more efficient energy management systems, contributing to the advancement of sustainable transportation solutions.

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